



Class: SY

Course: Fluid Mechanics

Course Code: PCC-CH202

Date: 25.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. This Paper Contains 2 Pages

Q.N.	Questions	Marks	CO	BT
Q.1	Provide the detailed classification of fluid with rheological equations and detail examples	(10)	CO1	BT3
Q.2	Determine the mass flow rate, mass velocity and the volumetric flow rate of the fluid flowing in a pipe of 40mm inside diameter, velocity of the water flowing through the pipe is 0.5 m/s, and density of water is 1000 kg/m ³	(10)	CO1	BT4
Q.3	What is friction factor? Explain the relationship between Stanton and Pannell friction factor, Moody's friction factor and Fanning (Darcy's) friction factor. With details diagram show the relationship between friction factor and Reynolds number. Also provide the various empirical relationships between Reynolds number and friction factor in laminar and turbulent zones.	(10)	CO2	BT5
Q.4	Provide the Bernoulli's Statement. Also derive the Bernoulli's Equation by considering frictional effect.	(10)	CO2	BT4
Q.5	What is the equivalent length of pipe? Three pipes of length 800m, 500m & 400m having diameters 500mm, 400mm and 300mm respectively are connected in series. These pipes are to be replaced by a single pipe of 1700m length. Find the diameter of single pipe. Use Dupuit equation.	(10)	CO3	BT4
Q.6	A 30 cm diameter pipe conveying water branches into two pipes of diameters 20 cm and 45 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s. Find the discharge in this pipe. Also determine the velocity in 15 cm pipe. If the average velocity in 20 cm diameter pipe is 2m/s.	(10)	CO3	BT5
Q.7	Derive the expression for Orifice meter for the mass flowrate flowing through the pipe. Also explain the term Vena Contracta.	(10)	CO4	BT5
Q.8	An aeroplane is moving horizontally in the sky and has its axis 5000 feet above the earth. The pitot tube properly placed just in-front of the plane and connected to modified manometer containing mercury. The difference in the mercury level	(10)	CO4	BT3

	is found to be 17 cms. Find the speed of the aeroplane. The density of mercury is 13600 kg/m^3 and density of air above 5000 feet is 1.056 kg/m^3 .			
Q.9	Explain construction working of centrifugal pump with neat labelled diagram also explain Pump cavitation b) priming of pump c) NPSH	(10)	CO5	BT3
Q.10	Write Short notes on (Any Two) a) Difference between rectangular and triangular notch b) Difference between orifice meter and venture meter (With diagram) c) Types of non-Newtonian fluids along with stress strain behaviour, rheological equations and suitable examples	(10)	CO5	BT3
Q.11	A conical pipe with smaller diameter in the upward position with the length of the pipe 3m. Water is flowing through the pipe where the velocity of the fluid is 6 m/s at the smaller diameter and the velocity is 2 m/s at the larger diameter pipe of the pipe. The loss of head in the pipe is given by the expression $0.2(u_1 - u_2)$. Where u_1 is the velocity of the fluid at smaller diameter and u_2 is velocity of the fluid at larger diameter. The pressure head is 3m of the fluid at smaller diameter. The flow of the fluid is downward direction. Determine the pressure head at the larger diameter. Density of the fluid is 1000 kg/m^3 . The diameter of the pipe is 20 cm of the smaller pipe cross-section. Determine the volumetric flow rate of water.	(10)	CO6	BT5
Q.12	A venture meter with throat diameter of 5cm is used to measure the flow rate of water flowing through the pipe of 15cm diameter. The reading of the manometer is 5cm of mercury. Determine the flow rate of water through the venture meter. The coefficient of discharge of venture meter is 0.98. The orifice meter of orifice diameter 5cm is used to measure the same flow rate of water in the pipe line. Determine the reading of differential manometer containing mercury under water. Coefficient of discharge of orifice meter is 0.6	(10)	CO6	BT5

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes (CO)

After Learning this Course Students will be able to

CO 1	To understand the concept of friction factor and analyze the fluid flow phenomena in various regions
CO 2	To understand the concept of friction factor and analyze the fluid flow phenomena in various regions
CO 3	To do the mass, momentum and energy balance and will be able to understand working of various flow meters.
CO 4	To understand and analyze and design the various flow-meters
CO 5	To understand and analyze various fluid transportation pumps
CO6	To understand and various types of pipe fitting, pumps and their performance characteristics.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Second Year

Course: Chemical Engineering
Thermodynamics-I

Course Code: PCC-CH203

Date: 21.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.

Q.N.	Question	Marks	CO	BT
Q.1	(a) Define state function and path function with two examples each.	3	1	1
	(b) A special manometer fluid has a specific gravity of 2.95 and is used to measure the pressure of 1.15 bar at a location where the barometric pressure is 760 mm Hg. Calculate the height of the manometer fluid in mm?	3	1	5
	(c) Derive an expression of the first law of thermodynamics for a flow process	4	2	3
Q.2	(a) State Gibb's phase rule and write a mathematical expression for it. A system consisting of some fluid is stirred in a tank. The rate of work done on the system by the stirrer is 2.25 hp. The heat generated due to stirring is dissipated to the surroundings. If the heat transferred to the surroundings is 3400 KJ/h, determine the change in internal energy.	5	2	5
Q.2	OR			
	(a) Iron filings are contained in a cylinder in an atmosphere of oxygen. It combines with oxygen according to the following reaction $2 Fe + \frac{3}{2} O_2 \rightarrow Fe_2O_3$ The pressure inside the cylinder is maintained at 101 kPa. The temperature is kept constant at 298 K by removing heat. For 2 mol of iron reacted, calculate Q, W, and ΔU. Given that 831.08 kJ of heat is liberated in the process	5	4	5
Q.3	(a) Classify the following properties into intensive and extensive properties. • Mole • Density • Specific Heat • Enthalpy	2	1	2&3
	(b) Provide clear distinction between the reversible and irreversible process in thermodynamics	3	1	2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Able to know the fundamental concept of thermodynamics.
CO 2	Able to apply first and second law of thermodynamics for chemical processes.
CO 3	Able to relate PVT relations and equation of state for pure component systems.
CO 4	Evaluate thermodynamic properties in chemical engineering applications.
CO 5	Understand the application of thermodynamics to various flow processes also of refrigeration and liquefaction



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Second Year

Course: Chemical Engineering
Thermodynamics-I

Course Code: PCC-CH203

Date: 21.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.

Q.N.	Question	Marks	CO	BT
	(a) Define state function and path function with two examples each.	3	1	1
Q.1	(b) A special manometer fluid has a specific gravity of 2.95 and is used to measure the pressure of 1.15 bar at a location where the barometric pressure is 760 mm Hg. Calculate the height of the manometer fluid in mm?	3	1	5
	(c) Derive an expression of the first law of thermodynamics for a flow process	4	2	3
Q.2	(a) State Gibb's phase rule and write a mathematical expression for it. A system consisting of some fluid is stirred in a tank. The rate of work done on the system by the stirrer is 2.25 hp. The heat generated due to stirring is dissipated to the surroundings. If the heat transferred to the surroundings is 3400 KJ/h, determine the change in internal energy.	5	2	5
Q.2	OR			
	(a) Iron filings are contained in a cylinder in an atmosphere of oxygen. It combines with oxygen according to the following reaction $2 Fe + \frac{3}{2} O_2 \rightarrow Fe_2O_3$ The pressure inside the cylinder is maintained at 101 kPa. The temperature is kept constant at 298 K by removing heat. For 2 mol of iron reacted, calculate Q, W, and ΔU. Given that 831.08 kJ of heat is liberated in the process	5	4	5
Q.3	(a) Classify the following properties into intensive and extensive properties. • Mole • Density • Specific Heat • Enthalpy	2	1	2&3
	(b) Provide clear distinction between the reversible and irreversible process in thermodynamics	3	1	2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Able to know the fundamental concept of thermodynamics.
CO 2	Able to apply first and second law of thermodynamics for chemical processes.
CO 3	Able to relate PVT relations and equation of state for pure component systems.
CO 4	Evaluate thermodynamic properties in chemical engineering applications.
CO 5	Understand the application of thermodynamics to various flow processes also of refrigeration and liquefaction